3873. Maximum Points Activated with One Addition

Solved

Hard Topics Hint

You are given a 2D integer array points, where points[i] = [x_i, y_i] represents the coordinates of the ith point. All coordinates in points are **distinct**.

If a point is **activated**, then all points that have the **same** x-coordinate **or** y-coordinate become **activated** as well.

Activation continues until no additional points can be activated.

You may add **one additional** point at any integer coordinate (x, y) not already present in points. Activation begins by **activating** this **newly added point**.

Return an integer denoting the **maximum** number of points that can be activated, including the newly added point.

Example 1:

> **Input:** points = [[1,1],[1,2],[2,2]]
>
> **Output:** 4
>
> **Explanation:**
>
> Adding and activating a point such as (1, 3) causes activations:
>
> - (1, 3) shares x = 1 with (1, 1) and (1, 2) -> (1, 1) and (1, 2) become activated.
> - (1, 2) shares y = 2 with (2, 2) -> (2, 2) becomes activated.
>
> Thus, the activated points are (1, 3), (1, 1), (1, 2), (2, 2), so 4 points in total. We can show this is the maximum activated.

Example 2:

> **Input:** points = [[2,2],[1,1],[3,3]]
>
> **Output:** 3
>
> **Explanation:**
>
> Adding and activating a point such as (1, 2) causes activations:
>
> - (1, 2) shares x = 1 with (1, 1) -> (1, 1) becomes activated.
> - (1, 2) shares y = 2 with (2, 2) -> (2, 2) becomes activated.
>
> Thus, the activated points are (1, 2), (1, 1), (2, 2), so 3 points in total. We can show this is the maximum activated.

Example 3:

> **Input:** points = [[2,3],[2,2],[1,1],[4,5]]
>
> **Output:** 4
>
> **Explanation:**
>
> Adding and activating a point such as (2, 1) causes activations:
>
> - (2, 1) shares x = 2 with (2, 3) and (2, 2) -> (2, 3) and (2, 2) become activated.
> - (2, 1) shares y = 1 with (1, 1) -> (1, 1) becomes activated.
>
> Thus, the activated points are (2, 1), (2, 3), (2, 2), (1, 1), so 4 points in total.

Constraints:

- 1 <= points.length <= 10^5
- points[i] = [x_i, y_i]
- -10^9 <= x_i, y_i <= 10^9
- points contains all **distinct** coordinates.

Seen this question in a real interview before? 1/5

Yes No

Accepted 6,556 /15.3K | Acceptance Rate 42.8 %

Topics

Hint 1

Hint 2

Hint 3

Hint 4

Discussion (23)

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